

Question 1 Random Forest: Full Trait Model

This notebook evaluates the custom CSC 466 random forest using `data/processed/breed_traits_full.csv`.

This version keeps each AKC trait rating as its own predictor, while `Coat Type` and `Coat Length` remain categorical.

Final Test Results

The model is trained on the training split using the best cross-validation parameters, then evaluated on the held-out test split. Macro F1 is useful here because the three popularity tiers are not perfectly balanced.

Model Interpretation

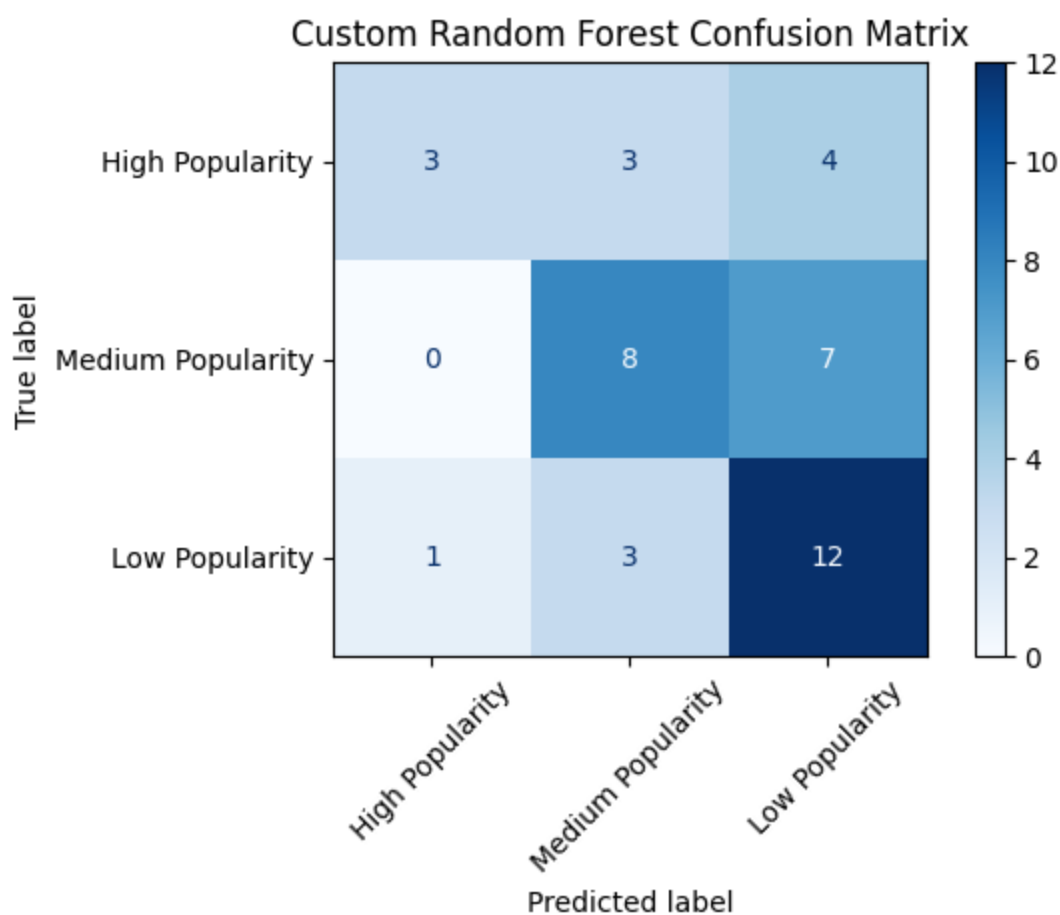
To interpret the model, these tables show which features the forest split on most often and how average trait values differ across the actual popularity tiers.

	Feature	Split Count
0	Barking Level	45
1	Watchdog/Protective Nature	39
2	Good With Other Dogs	32
3	Coat Grooming Frequency	28
4	Openness To Strangers	27
5	Shedding Level	26
6	Trainability Level	22
7	Energy Level	20
8	Affectionate With Family	17
9	Mental Stimulation Needs	16
10	Adaptability Level	16
11	Good With Young Children	12
12	Drooling Level	12
13	Playfulness Level	12

Popularity_Tier	High Popularity	Medium Popularity	Low Popularity
Affectionate With Family	4.65	4.46	4.47
Good With Young Children	4.37	3.70	3.73
Good With Other Dogs	3.82	3.29	3.62
Shedding Level	2.78	2.62	2.44
Coat Grooming Frequency	2.49	2.30	2.09
Drooling Level	1.90	1.88	1.58
Openness To Strangers	3.76	3.55	3.25
Playfulness Level	4.00	3.64	3.44
Watchdog/Protective Nature	3.76	3.92	3.49
Adaptability Level	4.08	3.71	3.66
Trainability Level	4.14	3.76	3.78
Energy Level	3.84	3.72	3.71
Barking Level	3.18	2.95	3.31
Mental Stimulation Needs	3.84	3.66	3.58

Coat Type	Coat Length
Popularity_Tier	
High Popularity	Smooth Short
Medium Popularity	Smooth Medium
Low Popularity	Smooth Short

	precision	recall	f1-score	support
High Popularity	0.75	0.30	0.43	10
Medium Popularity	0.57	0.53	0.55	15
Low Popularity	0.52	0.75	0.62	16
accuracy			0.56	41
macro avg	0.61	0.53	0.53	41
weighted avg	0.60	0.56	0.55	41



Prediction CSV

This output compares each test-set breed's actual popularity tier to the tier predicted by the full-trait random forest model.

	Breed	Actual Popularity Tier	Predicted Popularity Tier	Correct
0	Doberman Pinscher	High Popularity	Medium Popularity	False
1	Dogue de Bordeaux	Medium Popularity	Medium Popularity	True
2	Sussex Spaniel	Low Popularity	Low Popularity	True
3	Australian Cattle Dog	Medium Popularity	Low Popularity	False
4	Maltese	High Popularity	Low Popularity	False
5	Silky Terrier	Medium Popularity	Low Popularity	False
6	Yorkshire Terrier	High Popularity	High Popularity	True
7	Chinese Shar-Pei	Medium Popularity	Medium Popularity	True
8	Lakeland Terrier	Low Popularity	Low Popularity	True
9	Akita	Medium Popularity	Medium Popularity	True

10	Irish Wolfhound	Medium Popularity	Low Popularity	False
11	Bloodhound	Medium Popularity	Low Popularity	False
12	English Springer Spaniel	High Popularity	Medium Popularity	False
13	Greater Swiss Mountain Dog	Medium Popularity	Medium Popularity	True
14	Petit Basset Griffon Vendéen	Low Popularity	Low Popularity	True
15	Greyhound	Low Popularity	Low Popularity	True
16	American Eskimo Dog	Medium Popularity	Medium Popularity	True
17	Boston Terrier	High Popularity	Low Popularity	False
18	Pug	High Popularity	High Popularity	True
19	Harrier	Low Popularity	Low Popularity	True
20	German Wirehaired Pointer	Medium Popularity	Low Popularity	False
21	Afghan Hound	Medium Popularity	Low Popularity	False
22	German Shepherd Dog	High Popularity	High Popularity	True
23	American Water Spaniel	Low Popularity	Low Popularity	True
24	Pumi	Low Popularity	Low Popularity	True
25	Manchester Terrier	Low Popularity	Low Popularity	True
26	Azawakh	Low Popularity	Low Popularity	True
27	Neapolitan Mastiff	Medium Popularity	Medium Popularity	True
28	Norfolk Terrier	Low Popularity	Medium Popularity	False
29	Welsh Springer Spaniel	Low Popularity	Low Popularity	True
30	Chihuahua	High Popularity	Low Popularity	False
31	Bulldog	High Popularity	Low Popularity	False
32	Bearded Collie	Low Popularity	High Popularity	False
33	Shiba Inu	High Popularity	Medium Popularity	False
34	Redbone Coonhound	Low Popularity	Low Popularity	True
35	American English Coonhound	Low Popularity	Low Popularity	True
36	Otterhound	Low Popularity	Medium Popularity	False
37	Black Russian Terrier	Medium Popularity	Medium Popularity	True
38	Beauceron	Low Popularity	Medium Popularity	False
39	Schipperke	Medium Popularity	Low Popularity	False
40	Leonberger	Medium Popularity	Medium Popularity	True